

PFM 205 Course Outline

PFM 205 — Unit Cost and Service Norms

Course Outline

School	02 — Macroeconomics and Public Finance
Level	200 (analyst / practitioner)
Hours	30 (indicative band 24–32)
Shape	5 modules · 10 sessions · 6 laboratories · 4 assignments · 1 final assessment
Delivery	Instructor-led — online, blended or in person
Prerequisites	EDIS 101. PFM 204 is strongly recommended: this course maps its crosswalk against the classified budget PFM 204 produces, and inherits its analytical classification. PFM 203 is useful but not required. No prior Julia is assumed; only one laboratory needs a running instance, and it may be done on paper.
Assessment	Pass 60 %, with 60 % laboratory competency
Version	1.0 · reviewed 2026-08-24 · 12-month cycle

What this course is

PFM 205 teaches the AICTEM Unit Cost and Service Norms service: a 5,978-line Julia implementation across 13 layers, backed by a governed PostgreSQL store of 44 tables in 3 schemas with 54 CHECK constraints, 58 testsets and 412 assertions, and a generated formula reference of 179 entries.

UCSN holds the numbers that turn money into things: what a unit costs, what a unit yields, and what the operational factors between an approval and a commissioned asset are. It does not convert budgets — the Budget Output Engine does that, and PFM 206 teaches it. UCSN answers a resolve request with a value, its interval, its evidence tier, its provenance and its boundary — or with a refusal.

The rule this module states, and the one it borrows. The Budget module states the principle for fiscal data: no sector application should ever read the raw central budget workbook.

Who it is for

Sector planning and costing officers; budget analysts who are asked what a budget delivers rather than what it contains; monitoring and evaluation staff who must reconcile spending with output; statisticians

and data managers building an evidence base; and EDIS developers maintaining the parameter service or any module that consumes it.

What this course does not claim

This course does not convert budgets. The Budget Output Engine applies these coefficients, and PFM 206 teaches it.

Structure

Module	Title	Sessions	Laboratory	Hours
M1	What UCSN is; The parameter taxonomy and the layers	01–02	01	4.9
M2	The mathematics	03–04	02–03	7.4
M3	The four conversion modes; Evidence and governance	05–07	02–03, 06	6.6
M4	The utopia calibration	08–09	04–05	8.4
M5	Integration, errors and a worked dossier	10	04	1.7
ASM	Assessment and revision	—	—	1.0
	Total	10	6	30.0

The 10 sessions and 6 laboratories, in teaching order, and the subject matter of each:

M1 What UCSN is; The parameter taxonomy and the layers — Sessions 01–02 · 4.9 h · Laboratory 01

- **01 What UCSN is, and the rule that governs everything else** — the problem UCSN solves; two tiers, and the boundary between them; what UCSN is not; the rule that governs everything else, and more
- *Lab 01 Read the estate* — establish for yourself what UCSN is: how much code, what schemas, how many constraints, how many tests, and what the published outputs contain
- **02 The parameter taxonomy and the four layers** — ten parameter types; four layers; the line no coefficient may cross; the data model

M2 The mathematics — Sessions 03–04 · 7.4 h · Laboratory 02–03

- **03 The mathematics: identities and standardisation** — the core identities; standardisation
- *Lab 02 Convert a budget line by hand* — perform a full physical conversion with pencil and paper, including the bottleneck, and record every assumption
- **04 Estimation, uncertainty and quality** — robust estimation; uncertainty, and what to do with it; quality scoring; the resolver

- *Lab 03 Read an observation, and judge it* — decide whether a real observation from the Utopia store can support a coefficient, and say exactly what stops it

M3 The four conversion modes; Evidence and governance — Sessions 05–07 · 6.6 h · Laboratory 02–03, 06

- **05 The four conversion modes** — why four modes; the physical mode; the transactional and institutional modes; the sovereign mode
- **06 Execution, completion, leakage and the evidence base** — execution, completion and leakage; evidence tiers and epistemic classes; the observation store
- **07 Governance: approval, the rule, and validation** — versioning and approval; why no language model may author a coefficient; scenario overrides and isolation; validation rules
- *Lab 06 Break the governance, and watch it refuse* — establish by attempt, not by reading, that the controls hold

M4 The utopia calibration — Sessions 08–09 · 8.4 h · Laboratory 04–05

- **08 The Utopia calibration I — crosswalk and convertibility** — the budget crosswalk; convertible, pending and not convertible
- *Lab 05 Rank the coefficient gap* — turn the pending category into a costed acquisition programme
- **09 The Utopia calibration II — estimates and the gap** — the nine derived estimates; the illustrative set; ex-post unit expenditure; the evidence register, and more
- *Lab 04 Build a coefficient dossier* — produce the eight-section dossier of Chapter 37 for one real definition, and reach a recommendation

M5 Integration, errors and a worked dossier — Session 10 · 1.7 h · Laboratory 04

- **10 Integration, errors and reporting honestly** — UCSN and the Budget Output Engine; UCSN, the MTEF and FINEX; ten common errors; A worked coefficient dossier, and more

What a graduate can do

1. Explain what a governed parameter service is, and why no consumer of it may hold a coefficient of its own.
2. State the refusal rule and defend it against three arguments for a plausible default.
3. Assign a parameter to its layer, and explain why Layer C is registered rather than estimated.
4. Trace a value from observation through definition and estimate to an approved version.
5. Write the seven core identities and apply them to a real budget line.
6. Apply the six standardisation adjustments in the correct order, and quantify what applying them out of order costs.
7. Select an estimation method for a given evidence situation, and justify a robust estimator.
8. Read an interval with its semantics, and invert it correctly under division.
9. Perform a conversion in each of the four modes, and name the binding constraint in the physical case.
10. Distinguish execution, completion and leakage, and publish a residual as a residual.

The 16 formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	15 %	88 items across 10 modules
Laboratory exercises	30 %	6 laboratories, equally weighted, marked on method, interpretation, honesty about limits and communication
Applied assignments	25 %	convert a budget line and report it; judge five observations; build a coefficient dossier; rank the coefficient gap
Final assessment	30 %	Three hours, open book: definitions and structure; calculation; interpretation; reporting

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been recorded with its actual value: an unchecked result is no result.

Materials and set-up

A 162-page technical manual; 10 lecture decks and 6 laboratory decks; a laboratory manual, instructor guide and learner workbook; a 88-item quiz bank, 4 assignments and a final assessment with its marking guide; 8 datasets; 4 code exercises; 5 case studies; and the lms package. The README lists every file.

Every figure in this course is traceable to a published source or a committed model output; the sources are registered in `COURSE_VERSION.json` and the limitations in `COURSE_BUILD_REPORT.md`. Laboratories need only the extracts shipped in `08_Data/`.

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